

# Agentic Data Pipeline Orchestration for NASA Reanalyses — Colorado Snowpack

MERRA-2 and ERA5, Water Years 1981–2026

**Built with coding agents and agent skills**

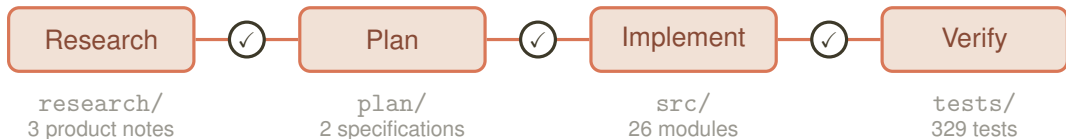
Responsible Gen-AI for NASA Earthdata Hackweek

[responsible-genai.hackweek.io](https://responsible-genai.hackweek.io)

August 28, 2026



# Project Planning



## Fixed Before Any Code

- the question
- domain and period
- variables, units, conversions
- the weighting rule
- what is out of scope

## Fixed Before Any Claim

- rank, never magnitude
- every number names its product
- peak SWE needs daily data
- never MERRA-2's 1 April SWE
- say where ERA5T is load-bearing

# Agent Skills

SKILL.md

name:

description: *when to use it*

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the traps

the conventions

what may be claimed

the agent matches on the description

... so it loads itself,  
before it can get it wrong



## Science

- reanalysis-snowpack-comparison
- snow-hydrology-fsca-evaluation
- modis-merra-regridding
- snow-bias-statistics-and-figures

## Run Integrity

- earthdata-streaming-checkpoints
- resumable-fetch-run-integrity
- derived-quantity-integrity

## Communication

- designing-thematic-figures

Eight skills, written during the work, from what went wrong in it.

# The Data Pipeline

**MERRA-2** Cloud OPeNDAP • earthaccess

**ERA5** ARCO-ERA5 Zarr • Google Cloud

**MODSCAG** NSIDC DAAC • Earthdata Login

**SNOTEL** NRCS AWDB REST • WTEQ

**3DEP** USGS elevation • staged once



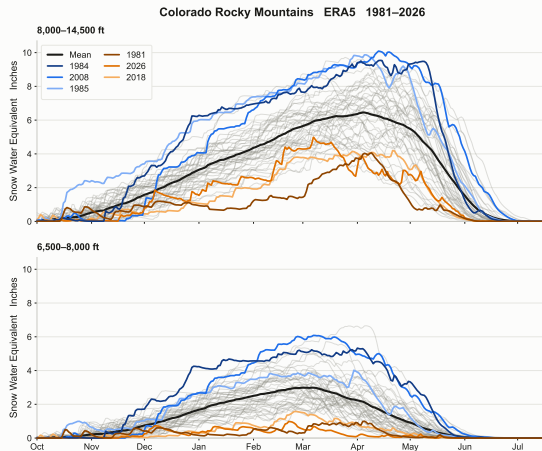
*resumable, and the whole report rebuilds  
from the checkpoints with no network*

## One Silent Trap per Archive

- **ERA5** *sd* is snow *water equivalent*, not depth. Geometric depth is  $sd / rsn$
- **MERRA-2** *SNODP* is depth *within* the snow-covered fraction; the grid mean is  $SNODP \times FRSNO$
- **ARCO-ERA5** the time axis runs to 2050. Absent chunks read back as NaN, with no error
- **SNOTEL** every station sits above the domain's median cell. Timing only, never magnitude

Each returns a plausible wrong number rather than an error. Each is now written down in a skill.

# ERA5 by Elevation Band



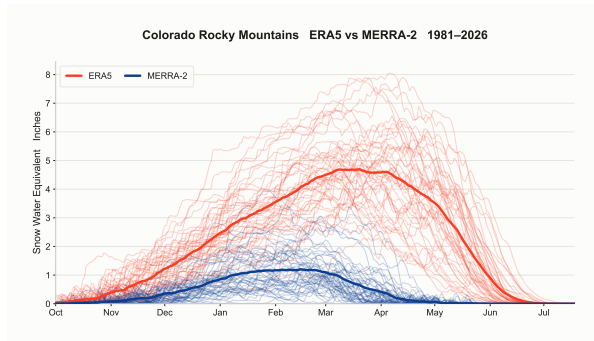
## 1 April SWE, per cent of band mean

|                 | 2023 | 2026 |
|-----------------|------|------|
| 8,000–14,500 ft | 140% | 51%  |
| 6,500–8,000 ft  | 297% | 5%   |

WY2026 is the lowest 1 April SWE of the 46 years, at 41% of the record mean. The deficit is not uniform: the lower band all but vanished.

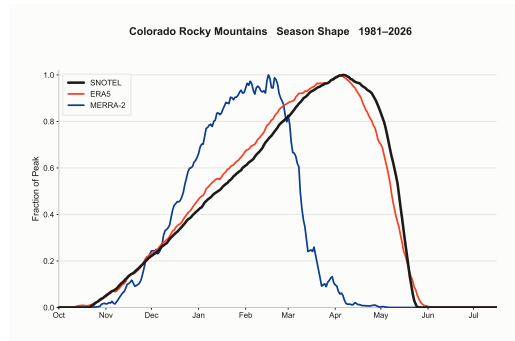
*ERA5. The two models disagree in magnitude, so every number names its product.*

# Two Reanalyses, One Observational Check



A factor of 3.3 apart in mean peak SWE.  
MERRA-2 exceeds ERA5's mean in 0 of 46 years.

*Yet they rank the years alike:  $\rho = 0.73$*

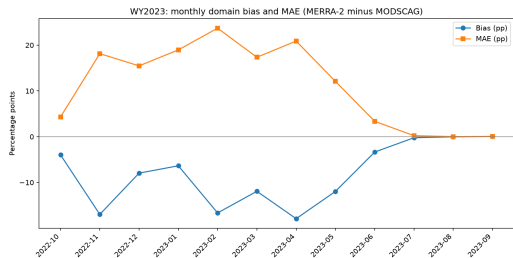
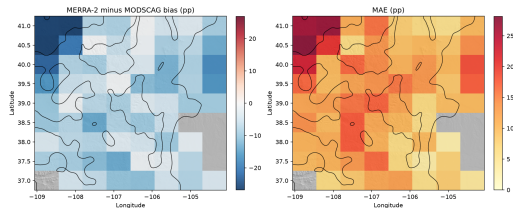
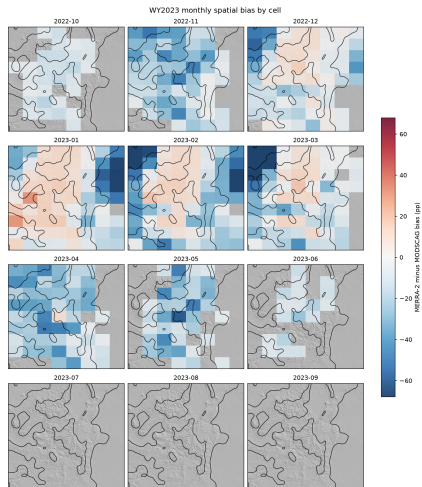


Each series scaled by its own peak.  
SNOTEL arbitrates timing only.

*MERRA-2 peaks two months early*



# fSCA Bias and MAE, Water Year 2023



MERRA-2 runs low against the satellite through accumulation and converges at melt-out.